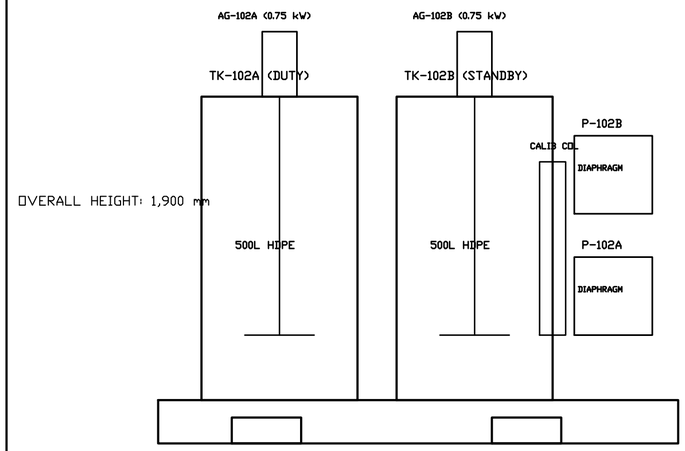


CHEMICAL REAGENT DOSING SKID SPECIFICATION & BILL OF MATERIALS (SK-100):

ITEM 01: Solution Preparation Tanks TK-102A/B (2 Units, Duty / Standby)

- Net Working Volume: 500 Liters each | Material: UV-Stabilized Food-Grade Cross-Linked HDPE (FDA 21 CFR 177.1520).
- Tank Geometry: 800 mm Dia x 1,100 mm Straight Shell Height | Conical bottom with 1.5" drain valve.
- Accessories: Hinged powder loading hatch with safety interlock, dust screen, ultrasonic level transmitter LT-102.

ITEM 02: High-Shear Agitator Mixers AG-102A/B (2 Units)

- Motor: 0.75 kW, 415V, 3-Phase, 50 Hz, IP55, TEFC | Shaft & Impeller: AISI SS316L 4-Blade Marine Propeller.
- Process Service: Rapid dissolution of food-grade Sodium Bicarbonate / Trisodium Citrate salts @ 25-35 deg C.

ITEM 03: Precision Metering Chemical Pumps P-102A/B (2 Units, Duty / Standby)

- Pump Type: Positive Displacement Mechanically Actuated PTFE Diaphragm Dosing Pumps.
- Flow Capacity: 5.0 to 50.0 L/h (Normal Operating Point: 14.06 L/h for 1.0 wt% reagent liquor recipe).
- Turndown: 10:1 Turndown ratio with micrometer stroke length adjuster + 4-20mA VFD speed modulation.
- Wetted Metallurgy: PVDF Liquid End, Ceramic Double Ball Check Valves, Pure Virgin PTFE Diaphragm.

ITEM 04: Skid Piping, Instrumentation & Safety Systems

- Process Piping: 1" NB Schedule 10S AISI SS316L piping with sanitary tri-clamp fittings.
- Pressure Relief Valves PSV-102A/B: Set @ 4.0 bar g with full recirculation bypass to suction tank.
- Pulsation Dampener: Bladder-type 0.5L PVDF dampener smoothing discharge pulsation to < +/-2.5%.
- Calibration Column: 1,000 mL Borosilicate glass cylinder with 3-way isolation valve for flow rate verification.
- Bund Containment: 110% Spill containment capacity (600L) with optical liquid leakage switch LS-102 interlocked to ESD.

SKID NOZZLE SCHEDULE & UTILITY INTERFACE SCHEDULE:

NOZZLE	SERVICE	SIZE	RATING	FACING	METALLURGY	CONNECTED TO
N1	Soft Water Supply Infeed	1.5" NB	ANSI 150#	Flat Face	AISI SS316L	WTP-01 Header @ 2.5 bar g
N2	Chemical Dosing Discharge	1.0" NB	ANSI 150#	Raised Face	AISI SS316L	Soak Battery TK-101 Feed Spool
N3	Containment Bund Drain	2.0" NB	ANSI 150#	Flat Face	AISI SS304	ETP Equalization Pit TK-104
E1	Main Electrical Power Feed	4-Core	415V / 50Hz	Glanded	Armoured Cu	MCC-01 Feeder Feat-04 (3.5 kW)
I1	Automation Signal Trunk	12-Pair	4-20mA / DI	Shielded	Tinned Cu	Remote I/O Panel RIO-02 (Modbus TCP)

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

EQUIPMENT GA - CHEMICAL DOSING SKID SK-100
DISCIPLINE: MECHANICAL | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-009	REV: V8.0	SHEET: 9 OF 13
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

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